



A Data-Driven Readiness Assessment Framework for Circular Economy and Industry 4.0 Adoption in Food and Beverage Small and Medium-Sized Industries in Developing Countries

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Abstract: Sustainable transformation within food and beverage (F&B) small and medium-sized industries (SMIs) in developing countries continues to be constrained by high levels of food loss and waste (FLW), inefficient resource utilization, limited technological capability, and weak organizational preparedness for digital and circular transition. Although the integration of Circular Economy (CE) principles and Industry 4.0 (I4.0) technologies has increasingly been recognized as a strategic pathway toward sustainable industrial development, limited empirical attention has been devoted to the assessment of organizational readiness for such transformation, particularly within resource-constrained SMIs. In response to this gap, a data-driven readiness assessment framework was developed to evaluate the preparedness of F&B SMIs for CE and I4.0 adoption. Survey data obtained from 150 F&B SMIs were analysed through a method integrating Principal Component Analysis (PCA), CRITIC method, and the TOPSIS. PCA was employed to identify latent readiness dimensions and reduce indicator redundancy, while the CRITIC method was utilized to derive objective indicator weights based on contrast intensity and inter-criteria conflict. Subsequently, the TOPSIS was applied to calculate composite readiness scores and classify firms according to their readiness levels. Eleven readiness dimensions were identified, among which sustainable sourcing and circular procurement, environmental value internalization, and human-centric managerial capability were found to exert the strongest influence on readiness performance. The findings further revealed that most participating SMIs were positioned within the developing readiness category, indicating that sustainability-oriented and digital transformation practices have been initiated but remain insufficiently institutionalized and operationally integrated. The results suggest that readiness for CE and I4.0 adoption is shaped not solely by technological infrastructure, but also by organizational culture, strategic procurement practices, managerial orientation, and workforce-related capabilities. The proposed framework contributes to the sustainability and industrial transformation literature by providing a robust and transferable decision-support instrument capable of supporting evidence-based managerial interventions and policy formulation aimed at accelerating sustainable industrial transition within F&B SMIs in developing economies.

Keywords: Circular economy; Industry 4.0; Open innovation; Readiness measurement; CRITIC; TOPSIS; Food and beverage SMEs

1 Introduction

Food loss and waste (FLW) have emerged as a critical sustainability challenge within global food systems, particularly in developing economies where inefficiencies in production, processing, and distribution remain prevalent. The food and beverage (F&B) industry contributes substantially to environmental degradation through high resource consumption, greenhouse gas emissions, and waste generation. As a result, it has become a key

focus of sustainability initiatives. Research shows that FLW is closely associated with resource inefficiencies across multiple stages of the food supply chain [1]. Additionally, studies have shown that food waste leads to significant economic losses and negative environmental impacts, such as increased greenhouse gas emissions [1, 2]. While large corporations have increasingly adopted advanced environmental management practices, small and medium-sized industries (SMIs)—which dominate the F&B sector in developing economies—often face structural, technological, and managerial constraints that hinder effective reduction of FLW [3, 4].

In response, circular economy (CE) principles have gained increasing attention as a framework for redesigning resource flows, reducing waste, and extending product life cycles. CE strategies, such as recovering resources and converting waste into value-added resources, offer considerable potential to reduce FLW while also improving environmental performance [2, 3]. Research shows that firms adopting circular business models significantly enhance sustainability performance. However, empirical evidence indicates that SMIs do not consistently adopt CE practices consistently or evenly, especially in developing countries where limited access to technology, capital, and knowledge constrains their effective implementation [4, 5]. This emphasizes that merely adopting CE principles is insufficient for achieving sustainable performance; additional enablers, such as access to technology and collaborative frameworks, are essential [3].

In conjunction with CE principles, Industry 4.0 (I4.0) technologies—such as the Internet of Things (IoT), big data analytics, and digital traceability systems—have significantly revolutionized environmental management by enabling real-time monitoring, predictive decision-making, and data-driven operational optimization [5]. In the F&B SMI sector, I4.0 can facilitate the identification of operational inefficiencies and reduce food waste. However, their implementation remains challenging in developing countries because there are substantial disparities in technological readiness and absorptive capacity [3, 5]. Research shows that I4.0 projects can improve sustainability practices when effectively implemented because they enhance firms' resource management capabilities [5, 6].

Recent literature suggests that open innovation (OI) can help SMIs overcome resource and capability constraints by enabling access to external knowledge, technologies, and partnerships [7]. Through collaboration with suppliers, customers, and research institutions, SMIs can access complementary capabilities that support the implementation of both CE and I4.0 practices [3, 8]. From an OI perspective, environmental practices are viewed as collaborative processes rather than individual pursuits, underscoring the importance of partnerships and ecosystems in advancing sustainable initiatives [3, 7]. Despite the theoretical framework, the practical role of OI in linking CE and I4.0 initiatives to environmental management outcomes, especially in reducing FLW, has not been sufficiently explored.

This study proposes an environmental management framework to examine how CE practices and I4.0 technologies support F&B SMIs in reducing FLW in developing countries. This study applies an integrated Principal Component Analysis (PCA)–CRITIC–TOPSIS approach to reduce dimensional complexity, assign objective weights, and evaluate SMI readiness for sustainable transformation. This study clarifies essential priorities for SMI managers and policymakers, providing actionable insights for enhancing FLW management in resource-limited contexts [5, 9].

Although previous studies have examined CE, I4.0, and OI as distinct domains, limited research has assessed the readiness of SMIs to integrate these paradigms in a holistic and operational manner, especially in developing countries. Existing research tends to focus on large firms, individual technologies, or isolated practices, resulting in fragmented insights. Moreover, the multidimensional nature of CE, I4.0, and OI introduces methodological challenges that are insufficiently addressed in prior empirical work, limiting the development of practical decision-support tools for SMIs and policymakers. Rather than focusing solely on adoption outcomes, this study conceptualizes readiness as a precursor to sustainable transformation. The proposed framework functions as a data-driven decision-support tool for identifying capability gaps, prioritizing managerial actions, and guiding policy interventions. It further assists engineering managers and policymakers in selecting capability-appropriate transformation strategies across procurement, workforce development, and digital infrastructure domains.

To address these gaps, this study proposes an OI-driven readiness measurement tool for F&B SMIs in developing countries. By integrating PCA for dimensionality reduction with multi-criteria decision-making techniques for readiness assessment, this research identifies latent readiness dimensions and evaluates SMIs' relative preparedness to adopt CE and I4.0 practices for effective FLW management. Rather than focusing only on adoption outcomes, this study measures readiness as a precursor to sustainable transformation. The proposed framework supports targeted managerial actions and policy interventions through data-driven assessment. In addition, the proposed readiness measurement framework is positioned as a decision-support tool that assists engineering managers and policymakers in selecting capability-appropriate transformation strategies and coordinating system-level sustainability interventions across procurement, workforce development, and digital infrastructure domains.

This paper contributes to sustainability research in three ways. First, it offers an integrative framework that brings together CE, I4.0, and OI within the context of F&B SMIs. Second, it develops a practical and objective readiness measurement tool that can support sustainability assessment and evidence-based decision-making. Third, it provides empirical insights that are relevant for managers and policymakers seeking to accelerate sustainability transitions, reduce FLW, and strengthen environmental management in developing-country SMIs. Accordingly, the

study responds to the need for actionable sustainability assessment tools and policy-relevant research, as emphasized by Challenges in Sustainability.

2 Literature Review

This section critically synthesizes the existing literature on CE implementation, I4.0-enabled environmental management, and OI mechanisms in F&B SMIs, with an overarching focus on sustainable performance in developing countries. While FLW reduction has frequently been addressed as a key operational challenge in the F&B sector, recent studies increasingly recognize it as an intermediate outcome rather than an ultimate objective [10, 11]. Accordingly, this review positions sustainable performance encompassing environmental, economic, and operational dimensions as the primary outcome of interest, and examines how CE practices, digital technologies, and OI collectively contribute to performance enhancement [12–14]. Section 2.1 reviews CE practices in the F&B sector and their implications for sustainable performance. Section 2.2 examines the role of I4.0 technologies in strengthening environmental management and operational efficiency. Section 2.3 discusses OI mechanisms in SMIs as enablers of sustainability-oriented transformation. Section 2.4 integrates these research streams by reviewing studies that combine CE, I4.0, and OI, particularly in the context of developing countries.

2.1 Circular Economy Practices and Sustainable Performance in the Food and Beverage Sector

The CE has been widely adopted as a strategic framework for enhancing sustainable performance in resource-intensive industries, including the F&B sector [15, 16]. By promoting closed-loop material flows, resource efficiency, and waste minimization, CE practices contribute to improved environmental performance while simultaneously generating economic and operational benefits [17, 18]. Prior studies highlight that practices such as by-product valorization, circular packaging, resource-efficient processing, and surplus redistribution can reduce material inputs, lower disposal costs, and improve overall process efficiency [19, 20]. These outcomes position CE not only as an environmental strategy but also as a performance-oriented management approach.

In the context of F&B SMIs, CE practices have been linked to enhanced cost efficiency, improved compliance with environmental regulations, and strengthened market competitiveness through sustainability differentiation [10, 15]. However, existing literature often reports uneven implementation outcomes, particularly in developing countries where SMIs face constraints related to capital, skills, and institutional support [21]. Moreover, many studies conceptualize CE adoption as a standalone operational initiative, without explicitly examining its systemic impact on sustainable performance dimensions or its interaction with technological and organizational enablers [22]. This indicates a need for more integrative perspectives that connect CE practices with broader management systems and innovation processes capable of sustaining performance improvements over time [12].

2.2 Industry 4.0 Technologies as Enablers of Environmental Management and Operational Efficiency

I4.0 technologies have transformed the way firms manage operations, resources, and environmental impacts by enabling real-time data collection, advanced analytics, and intelligent decision-making [23, 24]. In the F&B sector, technologies such as the IoT, big data analytics, artificial intelligence (AI), and blockchain facilitate enhanced traceability, predictive maintenance, and adaptive production planning [11, 25]. These capabilities support improved environmental management by reducing resource inefficiencies, emissions, and unnecessary waste, while simultaneously enhancing operational efficiency and responsiveness [26].

Empirical evidence suggests that digital monitoring and analytics can significantly improve energy efficiency, inventory turnover, and production accuracy, thereby contributing to both environmental and economic performance [27]. For SMIs, I4.0 adoption offers opportunities to overcome scale disadvantages through data-driven optimization [28]. Nevertheless, the literature also highlights substantial barriers, including high investment costs, limited digital capabilities, and uncertainty regarding return on investment challenges that are particularly pronounced in developing economies [29]. Furthermore, prior studies often assess I4.0 impacts primarily through productivity or technological lenses, with limited emphasis on its strategic role in driving sustainable performance [25]. This gap underscores the importance of positioning I4.0 as a core component of environmental management systems rather than merely a set of operational tools.

2.3 Open Innovation Mechanisms and Sustainability in Small and Medium-Sized Industries

OI has gained increasing attention as a mechanism through which SMIs can enhance their innovation capacity by leveraging external knowledge, resources, and partnerships [30]. In contrast to closed innovation models, OI emphasizes collaboration with suppliers, customers, research institutions, and digital platforms to co-create value and accelerate innovation outcomes [31]. For SMIs pursuing sustainability goals, OI offers a pathway to access advanced technologies, sustainability expertise, and best practices that are otherwise difficult to develop internally [32].

The literature indicates that OI can positively influence sustainable performance by enabling the co-development of eco-innovations, facilitating knowledge sharing related to environmental management, and supporting the adoption

of new business models [28, 33]. However, empirical research on OI in SMIs often focuses on innovation output or market performance, with sustainability outcomes treated as secondary effects [21]. Moreover, the specific role of OI in mediating or amplifying the impact of CE practices and I4.0 technologies on sustainable performance remains underexplored [34]. This suggests that OI should be examined not merely as a complementary strategy, but as a central organizational mechanism that integrates sustainability objectives with technological and operational transformation [14].

In SMI contexts, particularly in developing economies, OI readiness often depends on internal enabling conditions such as managerial commitment, employee capability development, and sustainability awareness, which support firms' ability to engage effectively in external knowledge exchange.

2.4 Integrating Circular Economy, Industry 4.0, and Open Innovation for Sustainable Performance in Developing Countries

An emerging body of literature calls for holistic frameworks that integrate CE principles, I4.0 technologies, and OI mechanisms to address complex sustainability challenges [13]. Within such frameworks, CE defines the sustainability-oriented performance goals, I4.0 provides the technological infrastructure for monitoring and optimization, and OI enables cross-organizational collaboration and knowledge exchange [27]. This integrative perspective is particularly relevant for F&B SMIs in developing countries, where fragmented supply chains and limited internal resources constrain isolated sustainability initiatives [22, 32].

Despite conceptual recognition of their complementarity, empirical studies that simultaneously examine CE, I4.0, and OI in relation to sustainable performance remain scarce [32]. Existing research often adopts partial integration approaches, focusing on pairwise relationships rather than capturing their combined and interactive effects. Furthermore, developing country contexts are frequently underrepresented, resulting in limited understanding of how institutional constraints, capability gaps, and market conditions shape sustainability outcomes in SMI-dominated sectors [32]. These gaps highlight the need for an OI-driven environmental management framework that integrates CE and I4.0 practices to enhance sustainable performance in F&B SMIs operating in developing economies.

Accordingly, the integration of PCA, CRITIC, and TOPSIS is designed to translate the conceptual linkage among CE, I4.0, and OI into an operational readiness assessment framework. PCA addresses the complexity of the multidimensional indicator set by identifying latent constructs that represent the underlying readiness dimensions. CRITIC subsequently provides objective weights for these dimensions by considering both contrast intensity and information conflict, reducing reliance on subjective judgments. TOPSIS then synthesizes the weighted dimensions into a relative readiness score by measuring each SMI's proximity to the ideal readiness profile. This sequential logic enables the study to move beyond identifying isolated adoption practices toward developing a structured decision-support tool that reveals which SMIs are more prepared, which readiness dimensions matter most, and where targeted managerial or policy interventions are required.

2.5 Research Gap and the Contribution of This Paper

Figure 1 presents the main literature map and identifies the key research gaps in integrating CE, I4.0, and OI practices in F&B SMIs. The novelty of this study lies in proposing an OI-driven readiness measurement tool for SMIs in developing countries by integrating PCA and TOPSIS. Several main novelties of this study are explained as follows.

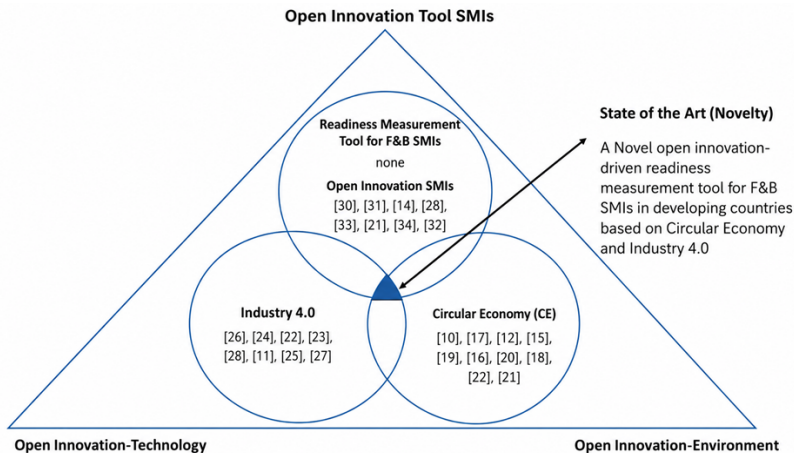


Figure 1. Summary of related works from pre-existing literature and the proposed research

Note: F&B SMIs = food and beverage small and medium-sized industries.

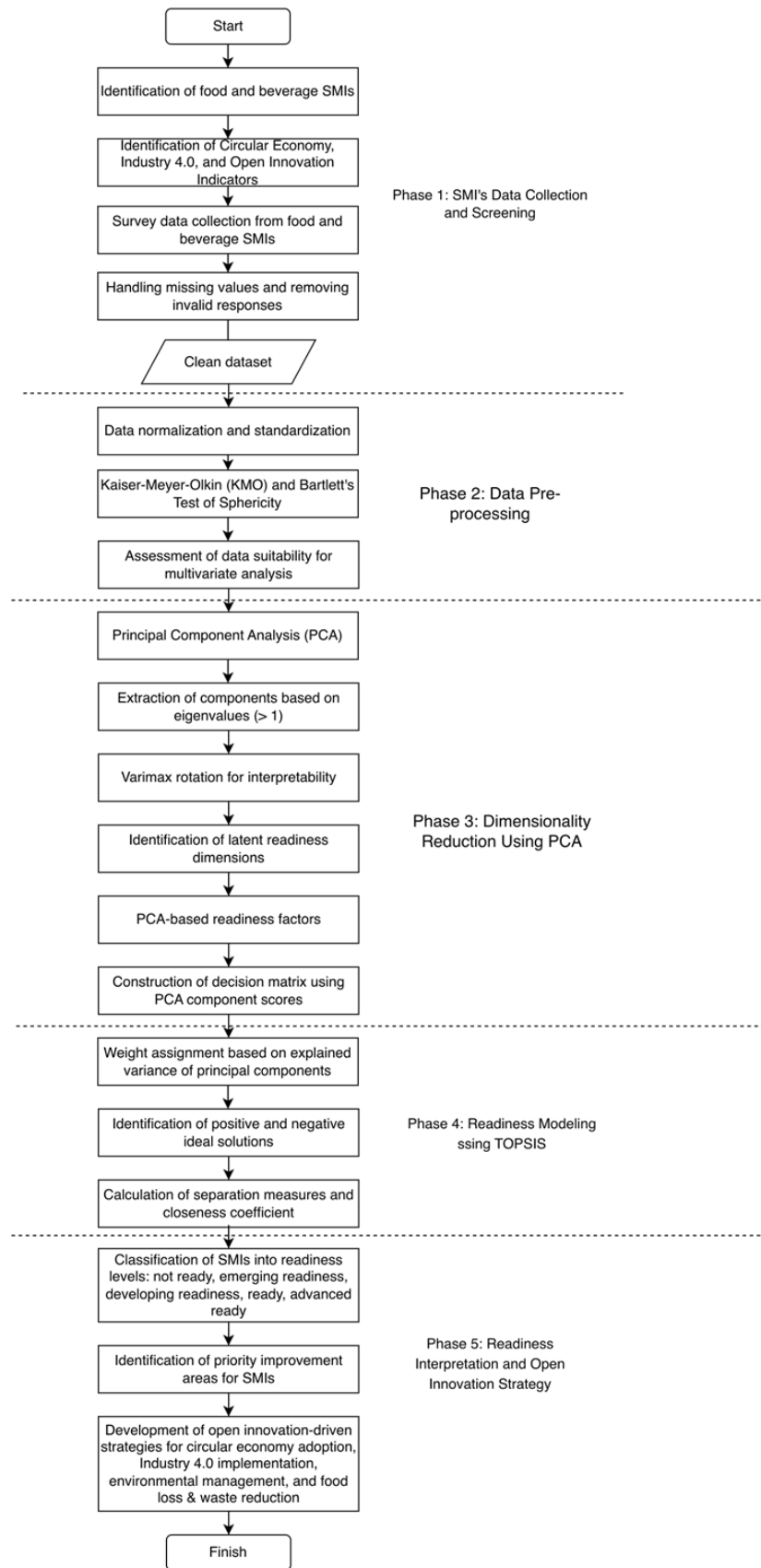


Figure 2. Overview of the proposed framework

Note: SMIs = small and medium-sized industries.

- First, this study is among the first to develop a readiness measurement framework for F&B SMIs by empirically integrating CE practices and I4.0 technologies from an OI perspective. Considering the limitations of previous studies, which often analyze CE, digitalization, or sustainability practices in isolation, existing research lacks a holistic readiness assessment that captures the multidimensional nature of SMI preparedness for sustainable transformation, particularly in developing-country contexts.

- Second, this study proposes a novel hybrid methodological approach combining PCA and TOPSIS to construct the readiness measurement tool. PCA is used to extract latent strategic dimensions from CE, I4.0, and OI indicators, addressing issues of multicollinearity and construct overlap commonly found in prior SMI sustainability studies. Subsequently, TOPSIS is employed to rank SMIs based on their relative readiness levels, enabling a systematic comparison of firms and supporting data-driven decision-making for sustainability-oriented transformation.

- Third, previous studies provide limited empirical guidance on how readiness for sustainable and digital transformation can be operationalized as a practical decision-support tool for SMI managers and policymakers. By integrating PCA-based dimensional reduction with TOPSIS-based ranking, this study moves beyond descriptive analysis and offers an actionable readiness measurement tool that identifies priority areas for improvement in environmental management and FLW reduction. Based on these research gaps, this study aims to support SMIs and policymakers in developing countries by providing an OI-driven, data-based approach to assess and enhance readiness for CE and I4.0 adoption.

3 Methodology

3.1 Research Procedure

This study proposes an OI-driven readiness measurement tool for F&B SMIs in a developing country context by integrating CE practices and I4.0 technologies within an environmental management perspective focused on FLW. An overview of the proposed research framework is illustrated in Figure 2. To assess readiness in a data-driven manner, this study employs a hybrid PCA and TOPSIS methodology, in which firm-level survey data are used to capture multidimensional indicators of CE, I4.0, and OI practices. This study is delineated into five primary stages.

In the first stage, data were collected from 150 F&B SMIs. The study focused on firms operating in Sleman Regency, Indonesia, as a representative developing economy context. The sample included firms from several F&B subsectors, such as processed food, beverage production, snack manufacturing, and traditional food businesses, thereby providing broader representation of the sectoral composition within the study area. A convenience sampling approach was employed, as the respondents were selected based on their accessibility and willingness to participate in the survey. Although convenience sampling limits statistical generalisation, it was considered appropriate for this exploratory readiness assessment because it enabled access to relevant firms within the target sector and local industrial context. The distribution of the questionnaire was facilitated through the database of the Department of Industry and Trade of Sleman Regency (Dinas Perindustrian dan Perdagangan Kabupaten Sleman), which helped identify and reach eligible firms within the study area. Firms included in the study were those classified as SMIs and actively operating in the F&B sector.

The second stage involved data pre-processing to eliminate incomplete responses and confirm the suitability of the dataset for subsequent PCA and TOPSIS analyses. In the third stage, PCA was applied to reduce the dimensional complexity of the observed variables and to extract latent strategic dimensions representing SMI readiness for adopting CE and I4.0 practices. This stage enables the identification of core readiness factors while addressing multicollinearity and overlap among indicators commonly found in sustainability and innovation-related variables.

In the fourth stage, the extracted principal components were integrated into a TOPSIS-based readiness modeling framework to assess the relative preparedness of the 150 SMIs. TOPSIS was employed to calculate the closeness coefficient of each SMI to the ideal readiness condition, allowing firms to be systematically ranked based on their readiness levels under an OI-driven environmental management perspective.

In the final stage, the PCA–TOPSIS results were interpreted to develop readiness categories and strategic implications for SMI managers and policymakers. This stage supports the identification of priority improvement areas related to environmental management and FLW reduction, thereby facilitating data-driven decision-making and targeted policy interventions for sustainable SMI development in developing countries.

While each method offers distinct analytical strengths, their standalone application is insufficient to capture the multidimensional complexity of SMI readiness for CE and I4.0 adoption. Figure 3 illustrates the integrated readiness measurement framework used in this study. The framework shows how CE, I4.0, and OI indicators are first consolidated into integrated readiness indicators and then processed through three sequential analytical stages: PCA, CRITIC, and TOPSIS. PCA effectively identifies latent readiness dimensions and reduces indicator redundancy; however, it does not provide a mechanism for evaluating or ranking SMI performance. In contrast, TOPSIS enables comparative assessment and ranking but relies on predefined weights, which are often subjectively assigned and may introduce bias. The CRITIC method addresses this limitation by generating objective weights based on data variability and informational conflict, ensuring that more discriminative dimensions receive greater importance.

Integrating PCA, CRITIC, and TOPSIS is necessary to establish a dimensionally robust, objectively weighted, and decision-oriented readiness assessment framework. This sequence also demonstrates how the proposed framework moves from multidimensional survey indicators to actionable readiness scores, rankings, and decision-support insights for managers, investors, and policymakers. This combined approach enhances methodological rigor and provides a more reliable and comprehensive evaluation than the use of each method in isolation, particularly in the context of heterogeneous SMI environments in developing countries.

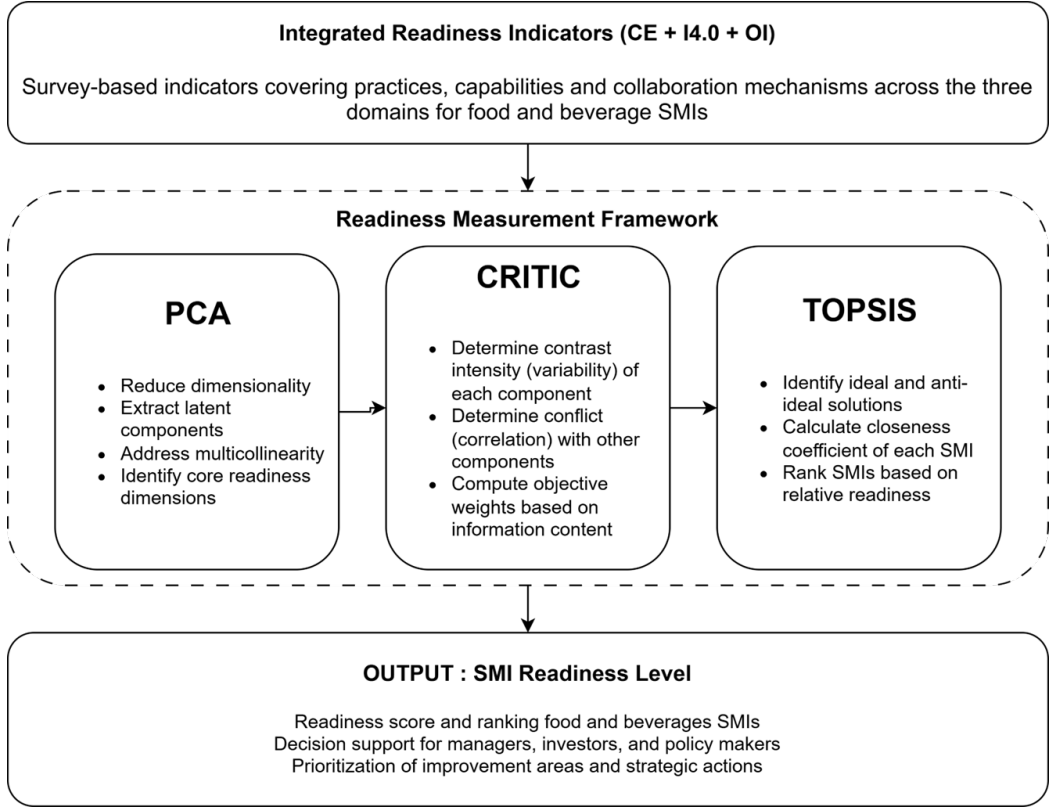


Figure 3. Integrated framework for OI-driven readiness measurement of F&B SMIs in developing countries
 Note: CE = Circular Economy; I4.0 = Industry 4.0; OI = open innovation; PCA = Principal Component Analysis; F&B SMI = food and beverages small and medium-sized industries.

3.2 Measurement of Variables

The variables used in this study were intended to capture organisational readiness rather than general organisational behaviour. In this context, readiness refers to the extent to which SMIs possess the capabilities, supporting mechanisms, and strategic orientation required to prepare for and implement sustainable and digital transformation. Accordingly, the variables represent three interrelated dimensions of readiness: CE, I4.0, and OI, as presented in Table 1, Table 2, and Table 3. All measurement items were adapted from validated scales in prior sustainability, innovation, and operations management studies, with contextual adjustments to reflect the operational conditions of F&B SMIs in developing countries. To further clarify construct validity, readiness in this study is treated as a multidimensional construct rather than a single technological or operational condition [35]. Therefore, the inclusion of heterogeneous indicators is intentional, as CE, I4.0, and OI readiness require different but interrelated capabilities. For I4.0, readiness includes both the availability of digital technologies and the ability to integrate them into operational processes [36]. For OI, readiness includes not only external collaboration, but also internal absorptive capacity, managerial support, and employee capability development [37, 38], which enable firms to access, assimilate, and apply external knowledge. This multidimensional interpretation is particularly relevant for F&B SMIs in developing economies, where sustainability and digital transformation often depend on the gradual development of internal capabilities before more advanced collaborative or technological practices can be implemented.

The CE variables reflect readiness because they indicate whether firms have established operational foundations for more resource-efficient and waste-reducing practices. These include resource efficiency, waste reduction, recycling and reuse, eco-design, and by-product utilisation. Such indicators show the firm's preparedness to redesign material flows and reduce FLW in line with circular production principles.

Table 1. Industry 4.0 (I4.0) independent variable

Code	Concept and Variable Industri 4.0	References
I1	IoT	[39]
I2	Cloud computing	[39]
I3	Online platform	[40–42]
I4	Big data	[40–42]
I5	Smart device	[40–42]
I6	AI	[40–42]
I7	Smart packaging	[40–42]
I8	Pemantauan real-time	[40–42]
I9	Cybersecurity	[40–42]
I10	Digital payment	[40–42]
I11	Mobile application adoption	[40–42]
I12	Digital marketing tools	[40–42]
I13	Supply chain digitalization	[39]
I14	Food safety technology	[39]

Table 2. 10R strategies (circular economy independent variable)

Strategy	Adaptive Description for Food & Beverage SMIs
R1–Refuse	Avoid the use of unnecessary food additives, such as excessive synthetic colorants or chemical preservatives, and limit single-use plastic packaging. Prioritize local, natural, and environmentally responsible ingredients.
R2–Rethink	Redesign products by considering optimal shelf life, nutritional value, and packaging that can be easily recycled. Introduce refill options or returnable container systems for customers.
R3–Reduce	Minimize food and packaging waste through better production planning, appropriate portion sizing, and the adoption of energy and water-efficient processing technologies.
R4–Reuse	Reuse safe edible by-products, such as fruit peels for jams or fermented products. Encourage customers to return glass bottles or containers for repeated use.
R5–Repair	Maintain and repair production equipment (mixers, ovens, freezers) to extend their service life and reduce unnecessary disposal. Reusable packaging items may also be repaired rather than replaced.
R6–Refurbish	Upgrade or modify small-scale production machinery to improve efficiency, such as replacing older motors with energy-saving alternatives.
R7–Remanufacture	Reprocess defective yet safe products into new items, for example converting misshapen bread into breadcrumbs, toppings, or fermented preparations.
R8–Repurpose	Convert organic residues into livestock feed, compost, or natural bio-enzymes. Vegetable scraps, for instance, can be used to create liquid organic fertilizer.
R9–Recycle	Recycle organic waste into compost or small-scale biogas. Used packaging materials can be transformed into handicrafts or other value-added items.
R10–Recover	Extract remaining value from production by-products (skins, seeds, pulps) to generate usable energy, such as small-scale bioenergy or briquettes made from coconut residue.

Note: SMIs = small and medium-sized industries.

The I4.0 variables operationalise readiness by capturing the firm's preparedness to adopt digital technologies and data-driven process management. These include digital monitoring systems, IoT-based tracking, data analytics, automation, digital traceability. In addition, supporting factors such as training and organisational incentives are treated as readiness indicators because they reflect workforce capability and internal support mechanisms that enable technological adoption, rather than merely routine managerial practices. In this study, OI is interpreted from a readiness perspective rather than a purely collaboration-based perspective. Accordingly, the OI indicators include both outward-facing collaboration mechanisms (e.g., partnerships with suppliers and stakeholders) and inward-oriented organisational enablers (e.g., environmental awareness, managerial support, and employee capability development). These internal elements represent absorptive capacity, which is widely recognised in the OI literature as a prerequisite for firms to effectively access and utilize external knowledge. This interpretation is particularly relevant for SMIs in developing-country contexts, where OI adoption typically evolves gradually from internal capability formation toward structured external collaboration. Therefore, the OI dimension reflects organisational openness to external knowledge rather than collaboration intensity alone.

The OI variables serve as indicators of readiness by assessing the firm's ability to access, absorb, and apply

external knowledge through collaboration. These include partnerships with suppliers and external actors, knowledge sharing, stakeholder co-creation, and participation in innovation networks. Such indicators represent the firm's absorptive capacity and openness to integrating external ideas and resources into internal operations.

Table 3. Open innovation (OI) independent variable

Code	OI Dimension	Measurement Items	References
P1	Personal capabilities	I have implemented environmentally friendly principles (such as waste reduction and energy efficiency) in my business operations.	[43]
P2	Personal capabilities	I routinely carry out reduce, reuse, and recycle (3R) activities in my business.	[44]
P3	Personal capabilities	I practice energy-saving measures in business operations (e.g., turning off electricity when not in use).	[45]
P4	Personal capabilities	I actively monitor market trends, particularly trends related to sustainability.	[46]
C1	Circular purchasing	I use raw materials that are recyclable or derived from natural resources.	[47]
C2	Circular purchasing	I use more environmentally friendly product packaging (e.g., reduced material or plastic usage).	[48]
C3	Circular purchasing	I collaborate with suppliers to acquire the knowledge and skills required for environmentally sustainable practices.	[49]
C4	Circular purchasing	We collaborate with partners (distributors, retailers, etc.) to reduce the environmental impact of our products.	[49]
M1	Managerial practices	I provide additional incentives or bonuses to employees or teams.	-
M2	Managerial practices	I engage in social activities or make donations from business revenues.	[50]
M3	Managerial practices	I offer flexible working hours to employees or team members.	-
M4	Managerial practices	I discuss work-related problems with employees or team members to find solutions.	-
M5	Managerial practices	I organize training programs for employees.	-
M6	Managerial practices	I have attended training related to occupational safety or workplace health.	-

3.3 Principal Component Analysis Implementation

All measurement items were derived from established scales in previous literature on sustainability, innovation, and operations management. To make sure the items were relevant to the situation, they were changed to better reflect the operational realities and resource limitations of small and medium-sized F&B businesses in developing countries. This adaptation improves content validity while keeping theoretical consistency, which lets the variables accurately reflect the many ways that SMIs are ready to adopt the CE and I4.0.

PCA was employed to identify latent strategic dimensions underlying the observed CE, I4.0, and OI variables and to mitigate multicollinearity among explanatory variables prior to regression analysis. The suitability of the data for PCA was assessed using:

- Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy, with values exceeding the recommended threshold of 0.60.

- Bartlett's Test of Sphericity, confirming statistically significant correlations among variables ($p < 0.05$).

Principal components were extracted using the eigenvalue-greater-than-one criterion and further interpreted through Varimax rotation to enhance factor interpretability. Variables with factor loadings of 0.50 or higher were retained, ensuring construct reliability and conceptual clarity. Each principal component was labeled based on the dominant characteristics of its constituent variables, representing distinct strategic dimensions of sustainable SMI practices.

Step 1. Data Preparation and Standardization

Because the indicators are measured on a Likert scale and may have different dispersions, the observed variables $X_1, X_2, \dots, X_p$ are standardized to z -scores. Eq. (1) shows that x_{ij} is the value of variable j for SMI i , $\bar{x}_j$ is the mean of variable j , and s_j is the standard deviation.

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j} \quad (1)$$

Step 2. Suitability Tests (KMO and Bartlett)

Before extracting components, sampling adequacy and correlation structure are assessed using KMO (acceptable if >0.60) and Bartlett's Test of Sphericity (significant if $p < 0.05$). These tests confirm that the correlation matrix is appropriate for PCA.

Step 3. Constructing the Correlation Matrix

PCA is performed on the correlation matrix of standardized variables. In Eq. (2), R denotes the correlation matrix, Z is the standardized data matrix, and n is the number of SMIs.

$$R = \frac{1}{n-1} Z^T Z \quad (2)$$

Step 4. Eigen Decomposition

Eigenvalues and eigenvectors are obtained by solving Eq. (3).

$$Re_k = \lambda_k e_k \quad (3)$$

where, λ_k is the eigenvalue of component k , e_k is the eigenvector (loading direction) for component k . Eigenvalues represent the amount of variance explained by each component.

Step 5. Component Retention Criteria

Components are retained using two actions. First, the Kaiser criterion retains components with $\lambda_k > 1$. Second, cumulative variance explained: retained components should explain an acceptable proportion of total variance ($\geq 60\%$).

Step 6. Factor Loadings and Rotation (Varimax)

Initial loadings are computed as Eq. (4). E is the matrix of eigenvectors and Λ is the diagonal matrix of eigenvalues. To improve interpretability, an orthogonal Varimax rotation is applied, producing rotated loadings L^* . Items with loading ≥ 0.50 are considered practically significant and used to interpret/label each component.

$$L = E\Lambda^{1/2} \quad (4)$$

Step 7. Computing Component (Factor) Scores

For each SMI i , the score of component k is computed as a weighted linear combination of standardized variables on Eq. (5). w_{jk} are the component score coefficients (derived from the PCA solution/rotation method). The set of PC_{ik} values represents the latent readiness dimensions for each SMI.

$$PC_{ik} = \sum_{j=1}^p w_{jk} z_{ij} \quad (5)$$

3.4 Development of the Readiness Measurement Tool (CRITIC-TOPSIS)

To quantitatively evaluate SMI readiness for adopting CE and I4.0 practices under an OI perspective, this study develops an integrated CRITIC-TOPSIS-based readiness measurement tool. The procedure consists of two sequential phases: objective weight determination using CRITIC and readiness scoring and ranking using TOPSIS.

Phase 1: Objective Weight Determination Using CRITIC

First, the initial decision matrix x_{ij} , where $i = 1, 2, \dots, m$ represents SMIs and $j = 1, 2, \dots, n$ represents readiness criteria, is normalized using the min-max normalization approach on Eq. (6).

$$r_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (6)$$

where, r_{ij} denotes the normalized value of criterion j for SMI i , while x_{ij} refers to the original value of criterion j for SMI i . The terms $\min(x_j)$ and $\max(x_j)$ represent the minimum and maximum values of criterion j across all SMIs, respectively. Here, m is the total number of SMIs and n is the total number of readiness criteria. Next, the contrast intensity of each criterion is calculated using its standard deviation, as shown in Eq. (7):

$$\sigma_j = \sqrt{\frac{1}{m} \sum_{i=1}^m (r_{ij} - \bar{r}_j)^2} \quad (7)$$

where, σ_j represents the standard deviation of the normalized values for criterion j , indicating the contrast intensity or discriminating power of that criterion across SMIs. The term r_{ij} denotes the normalized value of the criterion j

for SMI i , and $\bar{r}_j$ is the mean of the normalized values for criterion j across all SMIs. To assess redundancy among criteria, intercriteria correlation is measured using Pearson's correlation coefficient, as shown in Eq. (8):

$$c_{jk} = \rho_{jk} \quad (8)$$

where, c_{jk} denotes the correlation coefficient between criterion j and criterion k , while ρ_{jk} refers to the Pearson correlation coefficient between the two criteria. This correlation is used to determine the extent to which one criterion overlaps with another. The conflict measure for each criterion is then computed as shown in Eq. (9):

$$C_j = \sum_{k=1}^n (1 - c_{jk}) \quad (9)$$

where, C_j represents the conflict measure of criterion j , and c_{jk} denotes the correlation coefficient between criterion j and criterion k . A larger C_j value indicates that criterion j shares less redundancy with the other criteria and therefore contributes more unique information. The information content of each criterion is obtained by combining contrast intensity and conflict, as shown in Eq. (10):

$$S_j = \sigma_j \times C_j \quad (10)$$

where, S_j denotes the information content of criterion j , σ_j is the standard deviation of criterion j , and C_j is the conflict measure of criterion j . Thus, a criterion will have a higher information content when it both varies strongly across SMIs and has relatively low redundancy with other criteria. Finally, the objective weight of each criterion is calculated by normalizing the information content, as shown in Eq. (11):

$$w_j = \frac{S_j}{\sum_{j=1}^n S_j} \quad (11)$$

where, w_j represents the objective weight of criterion j , while S_j denotes the information content of criterion j . The denominator $\sum_{j=1}^n S_j$ is the total information content across all criteria. As a result, the weights are nonsubjective, sum to one, and reflect the relative importance of each readiness criterion.

Phase 2: Readiness Scoring and Ranking Using TOPSIS

Using the CRITIC-derived weights, the TOPSIS method is applied to compute the readiness score of each SMI. The decision matrix is first normalized using vector normalization on Eq. (12).

$$y_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m (x_{ij})^2}} \quad (12)$$

where, y_{ij} denotes the normalized value of criterion j for SMI i in the TOPSIS procedure, while x_{ij} is the original value of criterion j for SMI i . The term m refers to the total number of SMIs, and n refers to the total number of readiness criteria. The normalized matrix is then multiplied by the criterion weights to obtain the weighted normalized matrix, as shown in Eq. (13):

$$v_{ij} = w_j \times y_{ij} \quad (13)$$

where, v_{ij} denotes the weighted normalized value of criterion j for SMI i , w_j is the CRITIC-derived weight of criterion j , and y_{ij} is the normalized value of criterion j for SMI i . TOPSIS defines two reference points, namely the Positive Ideal Solution (PIS) and the Negative Ideal Solution (NIS), as shown in Eq. (14) and Eq. (15):

$$A^+ = \{\max(v_{ij}) \mid j = 1, 2, \dots, n\} \quad (14)$$

$$A^- = \{\min(v_{ij}) \mid j = 1, 2, \dots, n\} \quad (15)$$

where, A^+ denotes the PIS, which consists of the best value for each criterion. In Eq. (15), A^- denotes the negative ideal solution, which consists of the worst value for each criterion. The term v_{ij} refers to the weighted normalized value of criterion j for SMI i . The Euclidean distance of each SMI from the positive and negative ideal solutions is then calculated as shown in Eq. (16) and Eq. (17):

$$D_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - A_j^+)^2} \quad (16)$$

$$D_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - A_j^-)^2} \quad (17)$$

where, D_i^+ denotes the distance of SMI i from the PIS, whereas in Eq. (17), D_i^- denotes the distance of SMI i from the negative ideal solution. The term v_{ij} is the weighted normalized value of criterion j for SMI i , while A_j^+ and A_j^- represent the positive and negative ideal values for criterion j , respectively. Finally, the readiness score is expressed as the closeness coefficient, as shown in Eq. (18):

$$CC_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (18)$$

where, CC_i denotes the closeness coefficient or readiness score of SMI i , D_i^+ is the distance of SMI i from the PIS, and D_i^- is the distance from the negative ideal solution. The value of CC_i ranges from 0 to 1, where values closer to 1 indicate higher readiness.

$$CC_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (19)$$

Phase 3: Readiness Level Classification

To facilitate interpretation and strategic decision-making, SMIs are classified into readiness levels based on their closeness coefficient values in Table 4. The TOPSIS closeness coefficient (CC) ranges from 0 to 1. A higher CC value shows that an SMI is closer to the ideal readiness condition. TOPSIS generates a continuous ranking score and does not provide fixed readiness categories. Therefore, this study used an equal-interval classification method to make the results easier to interpret. The CC range of 0 to 1 was divided into five equal categories. These categories are Not Ready (0.00 to 0.20), Emerging Readiness (0.21 to 0.40), Developing Readiness (0.41 to 0.60), Ready (0.61 to 0.80), and Advanced Ready (0.81 to 1.00). This classification served as an interpretive framework to identify the relative readiness levels of the sampled SMIs [51].

Table 4. Readiness level classification

Closeness Coefficient (CC)	Readiness Category
0.00–0.20	Not Ready (NR)
0.21–0.40	Emerging Readiness (ER)
0.41–0.60	Developing Readiness (DR)
0.61–0.80	Ready (R)
0.81–1.00	Advanced Ready (AR)

This study adopted equal-interval thresholds because TOPSIS produces continuous relative performance scores without predefined categorical limits. Equal intervals allow consistent interpretation across firms and support comparison with previous readiness assessment studies. This method also fits exploratory decision-support research because it converts continuous ranking results into practical managerial categories. It does not aim to establish strict statistical cut-off points. Instead, the five readiness categories provide a clear structure for interpreting SMI readiness and supporting staged transformation planning.

4 Results

This section presents and discusses the empirical results of the proposed readiness measurement framework for F&B SMIs in developing countries. Following the analytical procedure outlined in the methodology, the results are organized into four main stages: data adequacy assessment, dimensionality reduction using PCA, objective weighting of the extracted readiness dimensions using the CRITIC method, and readiness scoring and classification through TOPSIS. This sequential analysis is intended to provide a comprehensive and data-driven understanding of how CE practices, I4.0 technologies, and OI mechanisms jointly shape SMI readiness for sustainable transformation. Beyond reporting the statistical results, this section also interprets their managerial and policy implications, particularly in relation to environmental management and FLW reduction.

4.1 Data Pre-Processing

In addition, the Bartlett's Test of Sphericity is statistically significant ($\chi^2 = 2,366.471$; $df = 703$; $p < 0.001$), rejecting the null hypothesis that the correlation matrix is an identity matrix. The results are presented in Table 5. This confirms that meaningful correlations exist among the variables, further validating the application of PCA. Overall,

the results of the KMO and Bartlett's tests provide strong empirical support for proceeding with PCA, indicating that the dataset is both adequate and reliable for extracting latent constructs related to OI-driven environmental management practices among F&B SMIs in developing countries.

Table 5. The result of Kaiser–Meyer–Olkin (KMO) and Bartlett's test

Test	Statistic	Value
KMO Measure of Sampling Adequacy	KMO	0.804
Bartlett's Test of Sphericity	Approx. Chi-Square	2,366.471
	Degrees of Freedom (<i>df</i>)	703
	Significance (Sig.)	<0.001

4.2 Dimensionality Reduction Using Principal Component Analysis

The results of the PCA (in Table 6 and Figure 4) indicate that 11 principal components were retained based on the eigenvalue-greater-than-one criterion (Kaiser's rule). Collectively, these components explain 66.48% of the total variance, which exceeds the commonly accepted threshold of 60% in social science and management research, indicating a satisfactory level of explanatory power.

Table 6. Eigen value of Principal Component Analysis (PCA)

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative (%)	Total	% of Variance	Cumulative (%)	Total	% of Variance	Cumulative (%)
1	8.853	23.298	23.298	8.853	23.298	23.298	3.507	9.228	9.228
2	3.254	8.563	31.861	3.254	8.563	31.861	3.467	9.123	18.351
3	2.418	6.363	38.224	2.418	6.363	38.224	3.110	8.185	26.536
4	1.792	4.716	42.940	1.792	4.716	42.940	2.749	7.234	33.770
5	1.680	4.422	47.362	1.680	4.422	47.362	2.356	6.201	39.971
6	1.477	3.886	51.248	1.477	3.886	51.248	2.330	6.130	46.102
7	1.370	3.606	54.854	1.370	3.606	54.854	2.203	5.797	51.899
8	1.202	3.163	58.017	1.202	3.163	58.017	1.578	4.153	56.052
9	1.143	3.008	61.025	1.143	3.008	61.025	1.403	3.692	59.745
10	1.052	2.768	63.793	1.052	2.768	63.793	1.341	3.528	63.273
11	1.021	2.688	66.481	1.021	2.688	66.481	1.219	3.208	66.481

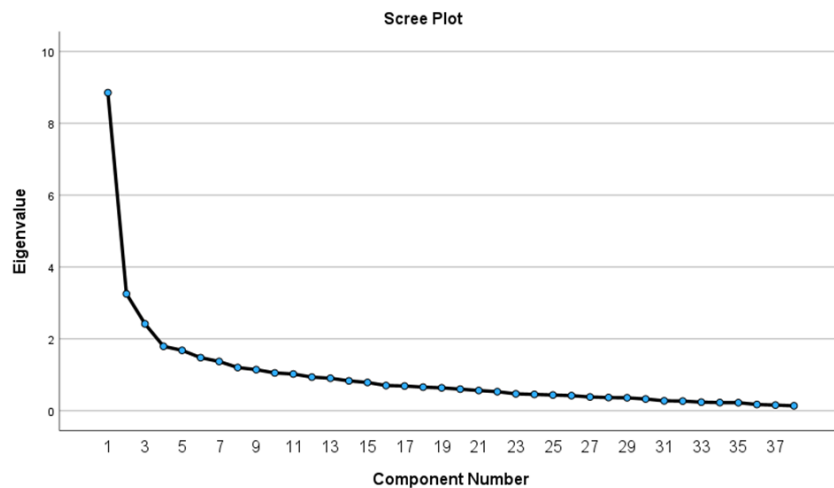


Figure 4. Scree plot of Principal Component Analysis (PCA)

The first principal component accounts for 23.30% of the total variance, suggesting the presence of a dominant latent dimension underlying OI-related practices among F&B SMIs. Subsequent components contribute progressively smaller yet meaningful proportions of variance, reflecting the multidimensional nature of OI, circular purchasing, and managerial practices embedded in SMI operations.

After Varimax rotation, the variance is more evenly distributed across the retained components. Each rotated component explains between 3.21% and 9.23% of the variance, enhancing interpretability by clearly separating

distinct strategic dimensions rather than concentrating explanatory power in a single factor. This balanced variance distribution supports a clearer theoretical labeling of each principal component in subsequent analysis.

Furthermore, the total variance explained confirms that the PCA effectively reduces the original set of observed variables into a manageable number of latent constructs while preserving most of the information contained in the data. These extracted components provide a robust foundation for the CRITIC–TOPSIS analysis, enabling the identification of key OI–driven factors influencing SMI success in environmental management and FLW reduction.

To enhance the interpretability and transparency of the PCA results, the rotated factor loadings were examined and summarised for each principal component. Only variables with loadings above 0.50 were retained and considered significant contributors to the corresponding component. Table 7 presents the main loadings for each component, highlighting the dominant indicators that form the basis for component interpretation and naming. Table 8 represents PCA correlation.

Table 7. Summary of main factor loadings (≥ 0.50)

PC	Indicators with High Loadings	Interpretation Basis
PC1	I10, I11, I13, I3	Digital infrastructure & platform integration
PC2	R7, R10, R6, R9, R4	Circular recovery & value retention
PC3	I6, I8, I7, I9	Intelligent & real-time digital systems
PC4	M6, M2, M5, I1	Managerial Learning, Social Responsibility, and Digital Coordination Capability
PC5	C3, C4, C2	Collaborative Circular Purchasing & Sustainable Supply Chain Integration
PC6	R2, R1, R3, R4	Circular Resource Optimization & Waste Prevention
PC7	P3, P4	Sustainability Awareness & Personal Adaptive Capability
PC8	P1	Environmental Value Internalization & Green Operational Mindset
PC9	R5	Asset Maintenance & Repair-Oriented Circular Capability
PC10	C1	Sustainable Sourcing & Circular Purchasing Capability
PC11	M3	Flexible Work Practices & Human-Centric Management

The naming of each principal component was therefore not assigned arbitrarily, but derived from the conceptual similarity and thematic consistency of the variables with the highest loadings within each component. For instance, PC1 is labelled as *Digital Infrastructure & Platform Integration Capability* because it is primarily defined by high loadings from digital payment (I10), mobile application adoption (I11), supply chain digitalization (I13), and online platform usage (I3), all of which reflect the firm’s digital operational backbone. Similarly, PC2 is interpreted as *Circular Recovery & Value Retention Capability* due to strong contributions from remanufacturing (R7), recovery (R10), refurbishing (R6), recycling (R9), and reuse (R4), which collectively represent downstream circular practices.

This approach ensures that the interpretation of principal components is grounded in empirical loading patterns rather than subjective judgement, thereby strengthening the robustness and validity of the PCA-based dimensional structure. Table 8 presents the correlation matrix among the 11 principal components extracted from the PCA. Overall, most inter-component correlations are low to moderate, indicating that the components represent related but distinct dimensions of readiness.

Table 8. Principal Component Analysis (PCA) correlation

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11
PC1	1.000	0.360	0.570	0.422	0.237	0.141	0.188	0.087	0.163	0.163	0.027
PC2	0.360	1.000	0.359	0.315	0.180	0.635	0.249	-0.056	0.395	0.240	-0.061
PC3	0.570	0.359	1.000	0.502	0.271	0.242	0.237	0.070	0.131	0.022	0.040
PC4	0.422	0.315	0.502	1.000	0.276	0.111	0.234	0.108	0.171	0.054	-0.056
PC5	0.237	0.180	0.271	0.276	1.000	0.185	0.169	0.155	0.200	-0.014	0.135
PC6	0.141	0.635	0.242	0.111	0.185	1.000	0.309	0.010	0.314	0.149	0.012
PC7	0.188	0.249	0.237	0.234	0.169	0.309	1.000	0.228	0.254	0.110	-0.014
PC8	0.087	-0.056	0.070	0.108	0.155	0.010	0.228	1.000	-0.024	0.006	-0.006
PC9	0.163	0.395	0.131	0.171	0.200	0.314	0.254	-0.024	1.000	0.084	0.025
PC10	0.163	0.240	0.022	0.054	-0.014	0.149	0.110	0.006	0.084	1.000	-0.050
PC11	0.027	-0.061	0.040	-0.056	0.135	0.012	-0.014	-0.006	0.025	-0.050	1.000

4.3 Weighting Results Using Multi-Criteria Decision Making (CRITIC)

Table 9 presents the results of the CRITIC-based objective weight calculation for the eleven readiness dimensions (PC1–PC11). The CRITIC method determines the relative importance of each principal component by jointly considering contrast intensity (σ_j) and conflict degree (C_j). Contrast intensity reflects the variability of each dimension across SMIs, while the conflict degree captures the extent to which a dimension provides unique information compared to others. The resulting objective weights (W_j) therefore represent the informational contribution of each readiness dimension to the overall readiness assessment.

Table 9. Weight calculation

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	PC11	Total	σ_j	C_j	W_j (Objective Weights)	%
PC1	0.000	0.640	0.430	0.578	0.763	0.859	0.812	0.913	0.837	0.837	0.973	7.640	0.268	2.051	0.073	7.28
PC2	0.640	0.000	0.641	0.685	0.820	0.365	0.751	1.056	0.605	0.760	1.061	7.384	0.276	2.041	0.073	7.25
PC3	0.430	0.641	0.000	0.498	0.729	0.758	0.763	0.930	0.869	0.978	0.960	7.555	0.287	2.165	0.077	7.69
PC4	0.578	0.685	0.498	0.000	0.724	0.889	0.766	0.892	0.829	0.946	1.056	7.864	0.239	1.881	0.067	6.68
PC5	0.763	0.820	0.729	0.724	0.000	0.815	0.831	0.845	0.800	1.014	0.865	8.206	0.347	2.846	0.101	10.11
PC6	0.859	0.365	0.758	0.889	0.815	0.000	0.691	0.990	0.686	0.851	0.988	7.891	0.258	2.032	0.072	7.22
PC7	0.812	0.751	0.763	0.766	0.831	0.691	0.000	0.772	0.746	0.890	1.014	8.036	0.239	1.918	0.068	6.81
PC8	0.913	1.056	0.930	0.892	0.845	0.990	0.772	0.000	1.024	0.994	1.006	9.421	0.357	3.359	0.119	11.93
PC9	0.837	0.605	0.869	0.829	0.800	0.686	0.746	1.024	0.000	0.916	0.975	8.286	0.296	2.452	0.087	8.71
PC10	0.837	0.760	0.978	0.946	1.014	0.851	0.890	0.994	0.916	0.000	1.050	9.236	0.460	4.247	0.151	15.08
PC11	0.973	1.061	0.960	1.056	0.865	0.988	1.014	1.006	0.975	1.050	0.000	9.948	0.318	3.164	0.112	11.24

The results indicate that PC10 (Sustainable Sourcing & Circular Purchasing Capability) has the highest weight (15.08%), followed by PC8 (Environmental Value Internalization & Green Operational Mindset, 11.93%) and PC11 (Flexible Work Practices & Human-Centric Management, 11.24%). The dominant weight of PC10 is driven by both its highest contrast intensity ($\sigma_j = 0.460$) and highest conflict degree ($C_j = 4.247$). This indicates that sustainable sourcing and circular purchasing practices vary substantially across SMIs and provide relatively unique information compared with other readiness dimensions. In practical terms, PC10 has strong discriminative power because some firms have already embedded circular procurement practices, while others remain at an early stage.

A similar pattern is observed for PC8, which records relatively high contrast intensity ($\sigma_j = 0.357$) and conflict degree ($C_j = 3.359$). This suggests that environmental value internalization differs markedly across firms and is not strongly redundant with other components. PC11 also receives a high weight, mainly due to its relatively high conflict value ($C_j = 3.164$), indicating that flexible work practices and human-centric management capture a distinct aspect of readiness. These findings suggest that upstream procurement decisions, embedded environmental values, and human-centric managerial practices play a dominant role in differentiating SMI readiness levels.

Moderate weights are observed for PC5 (Collaborative Circular Purchasing & Sustainable Supply Chain Integration, 10.11%) and PC9 (Asset Maintenance & Repair-Oriented Circular Capability, 8.71%), underscoring the relevance of inter-organizational collaboration and lifecycle extension practices in advancing CE readiness. In contrast, components such as PC4 (Managerial Learning, Social Responsibility, and Digital Coordination Capability, 6.68%) and PC7 (Sustainability Awareness & Personal Adaptive Capability, 6.81%) receive relatively lower weights. This is consistent with their lower σ_j and C_j values, indicating that although these dimensions are important enablers, they contribute less discriminative and less unique information in distinguishing readiness levels across SMIs.

The CRITIC results reveal that SMI readiness for CE and I4.0 adoption is primarily driven by upstream decision-making, embedded sustainability values, and human-centric management, rather than purely technological sophistication alone. By assigning objective, data-driven weights, the CRITIC method enhances the robustness of the readiness measurement tool and provides a solid foundation for the subsequent TOPSIS-based readiness scoring and ranking.

4.4 Readiness Scoring Results (TOPSIS)

Table 10 and Figure 5 summarise the readiness score distribution and readiness categories of the 150 F&B SMIs obtained through the integrated TOPSIS approach, while the detailed firm-level results are provided in Appendix A. In this study, the readiness score is expressed as the TOPSIS closeness coefficient (CC), calculated as Eq. (18). By construction, the closeness coefficient ranges from 0 to 1, where values closer to 1 indicate stronger readiness and values closer to 0 indicate lower readiness. No additional transformation or rescaling was applied to the reported CC values; therefore, the readiness scores presented in Table 10 represent the direct TOPSIS output.

As shown in Table 10, the majority of SMIs fall within the Developing Readiness category, accounting for 95 firms (63.33%), followed by the Ready category with 45 firms (30.00%). Only 10 firms (6.67%) are classified as Emerging Readiness, while no firms are included in the Not Ready or Advanced Ready categories. This distribution

indicates that most SMIs have moved beyond the initial stage of readiness, but have not yet reached a highly mature and fully integrated level of readiness.

Table 10. Readiness scoring results

Readiness Level	Score Range (CC_i)	Number of SMIs	Percentage
Not Ready (NR)	0.00–0.20	0	0.00%
Emerging Readiness (ER)	0.21–0.40	10	6.67%
Developing Readiness (DR)	0.41–0.60	95	63.33%
Ready (R)	0.61–0.80	45	30.00%
Advanced Ready (AR)	0.81–1.00	0	0.00%
Total		150	100.00%

Note: SMIs = small and medium-sized industries.

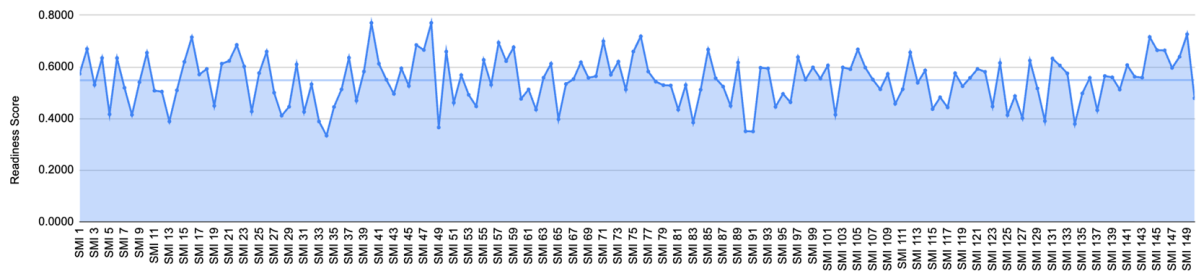


Figure 5. Readiness scoring for food and beverage (F&B) small and medium-sized industries (SMIs)

The dominance of firms in the 0.41–0.60 score range suggests that most SMIs have begun to adopt relevant practices associated with CE, I4.0, and OI, although such adoption remains partial and uneven. In practical terms, these firms may already demonstrate basic waste reduction efforts, initial use of digital tools, and some degree of external collaboration, yet their readiness is still constrained by limited integration across operational, technological, and organisational dimensions.

Meanwhile, the presence of 30.00% of firms in the Ready category indicates that a considerable proportion of SMIs have established stronger readiness foundations. These firms are more likely to show relatively consistent implementation of sustainability-oriented practices, wider use of digital technologies, and better managerial or organisational support for innovation. However, the absence of firms in the Advanced Ready category suggests that none of the sampled SMIs have yet achieved a fully mature readiness profile in which circular practices, digital capabilities, and collaborative innovation are comprehensively embedded in business operations.

Figure 5 further supports this interpretation by showing that most readiness scores cluster in the middle range, generally between 0.40 and 0.70, with moderate variation across firms and no extreme values near 0 or 1. Overall, the results suggest that F&B SMIs in Sleman Regency are in a transitional stage, where readiness has started to develop but still requires targeted support to progress toward higher levels of sustainable and digital transformation.

5 Discussion

5.1 Principal Component Analysis

The PCA effectively reduced the initial set of indicators into 11 significant principal components (PCs), with each component representing a distinct readiness dimension for F&B SMIs in adopting CE and I4.0 within an OI-driven environmental management context (Table 8). The extracted components demonstrate the multifaceted and systemic characteristics of SMI readiness, including internal competencies, digital facilitation, managerial oversight, collaboration, and sustainability-focused practices.

These components reflect the multifaceted and systemic nature of SMI readiness, including internal capabilities, digital enablement, managerial practices, collaboration, and sustainability-oriented activities. Importantly, these dimensions not only serve as measurement constructs but also provide a structured basis for decision-making by managers and policymakers when prioritizing transformation efforts.

This result is in line with prior studies on SMIs, which suggest that sustainable and digital transformation depends on the interaction of internal capabilities, technological preparedness, and external linkages rather than on a single factor [52].

5.1.1 PC1: Digital Infrastructure & Platform Integration Capability (I10, I11, I13, I3)

PC1 represents digital infrastructure and platform integration capability, reflecting the extent to which core business activities are supported by interconnected digital platforms. This dimension includes online platforms, digital payment systems, mobile applications, and digitally enabled supply chain processes.

From a readiness perspective, integrated digital systems improve operational efficiency, transaction transparency, and supply chain coordination. SMIs with higher scores on this component are better positioned to implement data-driven decision-making, enhance traceability, and support circular practices such as waste monitoring and resource optimization. In developing-country contexts, this capability acts as a key enabler for scaling sustainable operations and facilitating OI through digital connectivity.

From a decision-making perspective, SMIs with low scores should prioritize investments in basic digital infrastructure before advancing toward more sophisticated I4.0 technologies. Conversely, firms with high scores can leverage their digital maturity to implement real-time monitoring and advanced sustainability initiatives.

5.1.2 PC2: Circular Recovery & Value Retention Capability (R7, R10, R6, R9, R4)

PC2 represents SMIs' capability to retain value and recover resources through advanced CE practices, including reuse, refurbishing, remanufacturing, recycling, and recovery. These practices focus on extending material life cycles and minimizing waste.

From a readiness perspective, this dimension reflects the ability of SMIs to implement structured resource recovery systems beyond basic waste reduction. These practices often require coordination, technical know-how, and collaboration with external partners. SMIs with higher scores are better positioned to reduce food loss, improve environmental performance, and integrate circular and digital practices more effectively.

In terms of managerial implications, firms with low scores should prioritize partnerships for recycling and resource recovery, while policymakers can support this through industrial symbiosis programs and waste exchange platforms.

5.1.3 PC3: Intelligent Systems & Secure Real-Time Information Integration (I6, I7, I8, I9)

PC3 represents SMIs' adoption of intelligent digital systems and secure real-time information integration, reflecting advanced I4.0 capabilities that enable data-driven environmental management. This dimension includes artificial intelligence, smart packaging, real-time monitoring, and cybersecurity systems. From a readiness perspective, SMIs can leverage these technologies to improve traceability, predict inefficiencies, and reduce FLW. The integration of intelligent systems with cybersecurity safeguards ensures data reliability and operational continuity. SMIs scoring high on this component are better positioned to implement advanced, data-driven, and OI-oriented solutions that enhance both sustainability and competitiveness.

5.1.4 PC4: Managerial Learning, Social Responsibility, and Digital Coordination Capability (M6, M2, M5, I1)

PC4 represents SMIs' managerial capability to foster organizational learning, social responsibility, and digitally supported coordination. From a readiness perspective, this dimension highlights the importance of managerial learning and governance structures in supporting sustainable and digital transformation. Training activities enhance employees' competencies, while social responsibility practices strengthen organizational legitimacy and stakeholder trust. SMIs scoring high on this component demonstrate stronger managerial readiness to integrate CE and I4.0 practices.

This finding indicates that managerial capability development, particularly in training, organizational learning, and digital coordination, should be treated as a priority intervention area. This is especially important for SMIs that aim to scale sustainable practices but lack internal human resource readiness.

5.1.5 PC5: Collaborative Circular Purchasing & Sustainable Supply Chain Integration (C3, C4, C2)

PC5 captures SMIs' capability to implement circular purchasing through collaborative supply chain relationships. This dimension reflects a network-based approach, where sustainability outcomes are achieved through coordination across the value chain rather than internal actions alone. From a readiness perspective, collaboration with suppliers enables access to sustainability knowledge, technologies, and shared solutions.

From a decision-support standpoint, procurement and supplier collaboration should be prioritized as early-stage interventions, as they enable faster integration of circular practices compared to capital-intensive technologies.

5.1.6 PC6: Circular Resource Optimization & Waste Prevention (R1, R2, R3, R4)

PC6 represents SMIs' capability to optimize resource use and prevent waste generation through the CE hierarchy. Unlike downstream practices, this component emphasizes upstream and preventive actions. Refuse reflects avoiding unnecessary resource use, rethink captures strategic redesign, reduce focuses on minimizing inputs, and reuse extends product life. Together, these practices form a proactive approach to minimizing waste before it occurs.

5.1.7 PC7: Sustainability Awareness & Personal Adaptive Capability (P3, P4)

PC7 represents SMIs' personal-level capabilities related to sustainability awareness and adaptive behavior, reflecting the extent to which owners or key decision-makers actively engage in environmentally responsible actions and continuously monitor sustainability-related market trends. This component captures the human and cognitive foundation of readiness for CE and I4.0 adoption. The inclusion of energy-saving practices (P3) indicates a behavioral commitment to resource efficiency in daily operations, such as reducing unnecessary energy consumption. Meanwhile, monitoring sustainability-oriented market trends (P4) reflects an outward-looking and adaptive mindset, where SMI decision-makers stay informed about evolving environmental expectations, regulations, and customer preferences.

From a readiness perspective, PC7 highlights that sustainable and digital transformation in SMIs is not driven solely by technology or formal systems, but also by individual awareness, attitudes, and proactive learning behaviors. SMIs scoring high on this component demonstrate a stronger capacity to anticipate changes, align operational practices with sustainability trends, and respond effectively to external pressures. Furthermore, PC7 serves as an important enabler of OI, as sustainability-aware and adaptive individuals are more likely to seek external knowledge, adopt new ideas, and collaborate with stakeholders to improve environmental performance. In developing-country contexts, where formal support systems may be limited, such personal capabilities play a critical role in initiating and sustaining transformation efforts.

This suggests that awareness-building programs and sustainability training targeting SMI owners are critical policy levers, particularly in developing countries where transformation is often driven by individual decision-makers rather than formal systems.

5.1.8 PC8: Environmental Value Internalization & Green Operational Mindset (P1)

PC8 represents the extent to which SMIs have internalized environmental values into their core business operations, reflecting a green operational mindset at the personal and organizational levels. This component captures whether environmentally friendly principles—such as waste reduction and energy efficiency—are not merely acknowledged but actively embedded in daily business practices. The presence of a single dominant indicator suggests that environmental value internalization functions as a foundational readiness dimension, rather than a complex or multi-faceted capability. SMIs scoring high on this component demonstrate a clear commitment to sustainability principles that guide operational decisions and routines.

From a readiness perspective, PC8 serves as a baseline enabler for more advanced CE and I4.0 practices. Without the internalization of environmental values, investments in digital technologies or circular systems are unlikely to be sustained or effectively utilized. This component therefore reflects the normative and behavioral foundation upon which higher-order sustainability and innovation capabilities are built. Moreover, PC8 supports OI by shaping organizational openness toward environmentally oriented ideas, collaborations, and practices. SMIs that embed green values internally are more receptive to external knowledge and partnerships that promote sustainability, particularly in resource-constrained developing-country contexts.

5.1.9 PC9: Asset Maintenance & Repair-Oriented Circular Capability (R5)

PC9 represents asset maintenance and repair capability. This dimension emphasizes extending equipment life cycles to reduce waste and resource consumption. From a readiness perspective, SMIs with strong maintenance practices can achieve cost efficiency and environmental benefits simultaneously. When combined with digital monitoring, this capability supports more resilient and sustainable production systems.

5.1.10 PC10: Asset Maintenance & Repair-Oriented Circular Capability (R5)

PC10 represents SMIs' capability to implement sustainable sourcing through circular purchasing practices, particularly by selecting raw materials that are recyclable or derived from natural and renewable resources. This dimension reflects an upstream CE readiness, where environmental considerations are embedded at the procurement stage before production activities take place. By prioritizing recyclable and natural inputs, SMIs can reduce environmental impacts across the product life cycle and mitigate downstream FLW. From a readiness perspective, PC10 highlights the strategic importance of procurement decisions in enabling broader CE and I4.0 adoption. Sustainable sourcing often requires coordination with suppliers, traceability of material origins, and compliance with environmental standards—capabilities that can be strengthened through OI and digital technologies. SMIs with higher scores on this component are therefore better positioned to integrate circular practices across their supply chains and support long-term environmental management objectives in developing-country contexts. This finding implies that procurement decisions should be treated as a strategic entry point for circular transformation, where managers can initiate sustainability improvements without requiring substantial technological investment. Specifically, the proposed framework is intended to support decision-making related to investment prioritization, sequencing of I4.0 adoption, and identification of critical capability gaps in SMIs.

5.1.11 PC11: Flexible Work Practices & Human-Centric Management (M3)

PC11 reflects flexible work practices as part of human-centric management. In practical terms, this component shows whether a firm provides adaptable working arrangements and creates a supportive organisational climate for employees. Firms with higher scores on this component are more likely to have a workforce that is motivated, open to change, and better able to adapt to new practices. This means that PC11 reflects an important soft dimension of readiness, where people management supports the firm's broader sustainability and digital transformation efforts.

Taken together, the identified principal components represent not only distinct readiness dimensions but also interconnected elements within a broader operational system. These dimensions are closely linked to system-level performance in F&B SMIs. Digital infrastructure influences efficiency, while supply chain collaboration affects upstream and downstream performance. These findings demonstrate that improvements in readiness directly contribute to production and supply chain performance, aligning the framework with an engineering management and systems-thinking perspective.

5.2 Readiness Scoring Analysis

A considerable proportion of SMIs are classified as Ready (CC between 0.61 and 0.80), indicating relatively strong preparedness for sustainable and digital transformation. These SMIs perform well across key readiness dimensions, such as sustainable sourcing, environmental value internalization, human-centric management, and digital infrastructure, highlighting their ability to translate sustainability awareness into operational practices. In contrast, a smaller group of SMIs remains in the Emerging Readiness category (CC between 0.21 and 0.40), indicating limited adoption of CE and I4.0 practices as well as weaker organizational readiness.

No SMIs fall into the Not Ready category, suggesting that all SMIs demonstrate at least a minimum level of awareness or engagement with sustainability and digitalization issues. This finding reflects the growing diffusion of CE and I4.0 concepts among F&B SMIs in developing countries, although at varying levels of maturity.

Overall, the readiness distribution highlights the heterogeneous nature of SMI preparedness and reinforces the need for differentiated managerial strategies and policy support tailored to specific readiness levels. This indicates that policy interventions should go beyond basic awareness-building and focus on accelerating capability development and bridging the gap between initial adoption and full integration of circular and digital practices.

To further support decision-making, the results can be operationalized into a priority-action framework based on the combination of dimension weights and readiness scores. High-weight dimensions with relatively low performance should be prioritized, as improvements in these areas are expected to generate the greatest impact on overall readiness.

For instance, if SMIs exhibit low performance in human-centric management or sustainable sourcing—identified as highly influential dimensions—targeted investments in workforce development and procurement restructuring should be prioritized.

Table 11 shows that the average readiness scores across industry categories generally remain within the Developing Readiness range, indicating that moderate readiness is the dominant pattern across the F&B sector. This suggests that the readiness gap is not limited to one specific product category, but reflects a broader structural issue affecting multiple types of SMIs. This interpretation is consistent with prior studies showing that SMIs often adopt CE practices unevenly across different business functions, while digital transformation in agri-food firms is also shaped by organisational and contextual determinants rather than by technology alone [53]. Overall, most categories fall within the Developing Readiness range (0.41–0.60), indicating that SMIs have begun adopting sustainability and digital practices but have not yet fully integrated CE and I4.0 approaches.

Table 11. Readiness small and medium-sized industries (SMIs) classification

Industry Classification	Code	Number of SMIs	Average Readiness Score
Other/General F&B Products	1	4	0.6112
Bakery and Pastry Products	2	12	0.5333
Agriculture-, Livestock-, and Fishery-Based Products	3	55	0.5512
Traditional Food Products	4	29	0.5437
Beverage and Herbal Products	5	5	0.5438
Ready-to-Eat Processed Food Products	6	40	0.5402
Snack and Confectionery Products	7	5	0.5824

The Other/General F&B Products category records the highest average readiness score (0.6112), placing it at the lower boundary of the Ready category. However, this result should be interpreted with caution because the group includes only four firms. Its relatively higher score may reflect greater operational flexibility and fewer process-specific constraints, making it easier to experiment with digital tools and sustainability-oriented practices. Their higher readiness may reflect greater operational flexibility and fewer process constraints. Similarly, Snack and

Confectionery Products show a relatively high readiness score (0.5824), which may indicate that more standardised production routines and less perishable inputs make it easier to adopt structured process improvements and basic digital coordination. In this sense, higher readiness in these categories may be associated with simpler implementation pathways for process control, traceability, and operational upgrading.

By contrast, categories with larger sample sizes, such as Agriculture-, Livestock-, and Fishery-Based Products, Traditional Food Products, and Ready-to-Eat Processed Food Products, record average readiness scores between 0.54 and 0.55. These categories remain within the Developing Readiness range. These results suggest that SMIs in these sectors face greater challenges, including supply chain complexity, perishability of raw materials, and the need for operational standardization. These factors may hinder the adoption of advanced circular and I4.0 practices. Similarly, the lower readiness observed in Bakery and Pastry Products suggests that more traditional or craft-oriented production systems may face difficulties in digital integration and circular resource management. Although these firms are not poorly prepared, they remain in the Developing Readiness stage, suggesting that transformation is still incomplete. A likely explanation is that these categories face more complex supply chains, stronger dependence on traceability, and greater sensitivity to raw material quality and perishability, all of which make the consistent implementation of advanced circular practices and digital systems more demanding. This explanation is supported by prior studies emphasizing that food supply chains are highly sensitive to traceability requirements and that increasing agri-food supply chain complexity heightens the need for reliable tracking and data systems [54].

The relatively lower score of Bakery and Pastry Products may also reflect the persistence of more craft-oriented production settings, where digitalisation and process automation can be harder to introduce consistently. This interpretation should be viewed as an inference from the present data, but it is broadly aligned with recent evidence showing that the bakery sector faces digital transformation challenges related to resistance to new technologies, high implementation costs, shortage of expertise, and concerns about preserving artisanal quality.

Overall, the readiness mapping reveals substantial heterogeneity among F&B SMIs. This variation suggests that readiness is shaped by product type, process complexity, and supply chain structure, rather than by technology adoption alone [53, 55]. These findings reinforce the need for tailored readiness strategies instead of applying a one-size-fits-all approach. Accordingly, policy interventions and managerial support should be designed based on the specific needs and operational characteristics of each F&B segment [55, 56].

More importantly, this study moves beyond descriptive assessment by providing a decision-oriented framework that links readiness measurement to actionable strategies. The proposed approach enables managers to identify priority intervention areas and supports policymakers in designing targeted programmes based on sectoral readiness levels. In this respect, the framework serves not only as an evaluation tool, but also as a practical decision-support system for staged transformation in resource-constrained SMI contexts [53, 56].

From a practical managerial perspective, the framework can be applied in a step-by-step manner. First, managers assess their firm's readiness using the proposed indicators and obtain a composite readiness score and classification. Second, they identify low-performing dimensions, particularly those with high importance weights, as priority areas for improvement. Third, resources can be allocated and targeted interventions can be designed based on the identified gaps. Finally, the framework can be applied iteratively to monitor progress and adjust strategies over time. This staged logic is consistent with prior studies showing that SMI transformation in sustainability and digitalisation is typically gradual rather than immediate [55].

From an engineering management perspective, the proposed framework supports system-level decision-making. It enables managers to allocate resources more effectively, prioritise high-impact interventions, and improve operational performance across production and supply chain systems. By linking readiness dimensions to performance outcomes such as efficiency, adaptability, and resource utilisation, the model facilitates not only assessment but also continuous system optimisation. This is aligned with earlier findings that CE and I4.0 adoption in SMIs depend on the integration of operational, managerial, and organisational capabilities rather than isolated technical improvements [53].

5.3 Managerial Implications

The findings of this study provide important managerial implications by translating the PCA–CRITIC–TOPSIS results into actionable decision-support insights for F&B SMIs. Rather than functioning solely as an assessment tool, the proposed framework enables managers and policymakers to prioritise interventions, allocate limited resources more effectively, and design staged transformation pathways toward CE and I4.0 adoption. This implication is consistent with prior empirical studies showing that CE and digital transformation in SMIs are not driven by technology alone, but also by organisational, managerial, and contextual factors that shape firms' ability to absorb change and sustain implementation over time.

First, the CRITIC-based weighting results provide a clearer basis for prioritisation. The prominence of dimensions such as sustainable sourcing and circular purchasing (PC10), environmental value internalisation (PC8), and human-centric management (PC11) suggests that managerial efforts should not focus exclusively on technological

upgrading. Instead, the findings indicate that readiness is strongly influenced by whether firms embed sustainability in procurement decisions, internalise environmental values in daily operations, and maintain workforce-related practices that support adaptation. This interpretation is in line with evidence that CE adoption in SMIs is closely related to design, resource efficiency, and broader organisational commitment, while I4.0 adoption is often constrained when firms overlook human capabilities, managerial support, and implementation readiness. In this sense, the results imply that upstream practices and organisational culture act as enabling conditions for more visible digital and operational transformation.

Second, the TOPSIS-based readiness classification supports a staged decision-making logic. Firms in the Emerging and Developing readiness categories should prioritise foundational capabilities, such as basic waste reduction routines, initial digital adoption, and stronger internal awareness of sustainability principles. By contrast, firms classified as Ready are in a better position to invest in system integration, including supply chain digitalisation, real-time monitoring, and collaboration with external partners. This staged interpretation is consistent with prior empirical work showing that SMIs and agri-food organisations tend to adopt digital transformation incrementally rather than through abrupt full-scale change, largely because readiness is shaped by differences in organisational maturity, infrastructure, and managerial capability. A stepwise strategy is therefore more realistic than a one-size-fits-all transformation model, particularly for resource-constrained SMIs.

Third, the identified principal components can be interpreted as interconnected subsystems within engineering management practice. The digital-related components point to the role of infrastructure, data integration, and intelligent monitoring in improving process visibility and operational control. The CE-related components highlight the importance of material efficiency, waste prevention, and value recovery in strengthening environmental performance. Meanwhile, the OI and managerial components reflect the significance of collaboration, learning, and human-centred coordination in sustaining transformation. At the system level, these three dimensions operate in a mutually reinforcing manner rather than as isolated capabilities. I4.0 enables real-time monitoring, digital integration, and data-driven operational control, while OI facilitates knowledge sharing, collaboration, and coordination across supply chain actors. These capabilities collectively support CE implementation by improving resource efficiency, waste reduction, and value recovery practices. Consequently, managerial decision-making in F&B SMIs increasingly depends on the ability to simultaneously align technological adoption, collaborative innovation, and sustainability-oriented operational strategies. Taken together, these results suggest that readiness is systemic rather than isolated: firms are unlikely to progress when only one subsystem improves while others remain weak. This helps explain why some firms may already adopt specific digital tools or environmental practices but still remain in intermediate readiness levels, because deeper integration across operations, people, and supply chains has not yet been achieved. This interpretation is broadly aligned with empirical studies showing that digital transformation and CE implementation depend on the interaction between internal capabilities, organisational support, and external linkages.

Fourth, the framework contributes to engineering management practice by positioning readiness assessment not only as a measurement exercise, but also as a structured decision-support mechanism for sustainability planning in SMI environments. From this perspective, the PCA results help managers identify which readiness dimensions are relatively strong or weak, the CRITIC weights indicate which dimensions deserve greater attention, and the TOPSIS ranking provides a practical basis for benchmarking firms and sequencing interventions. This integrated use of dimensional analysis, weighting, and ranking is especially valuable in F&B SMIs, where managerial decisions are often made under financial constraints, operational uncertainty, and limited access to advanced expertise. Empirical studies on SMI digitalisation and CE adoption similarly emphasise that firms benefit from practical prioritisation mechanisms because transformation resources are rarely sufficient to improve all dimensions simultaneously.

Finally, the proposed framework also has implications for policymakers and support institutions. By identifying dominant readiness dimensions and classifying firms into different readiness levels, public support can be targeted more precisely. For lower-readiness firms, training, awareness-building, and basic capability development may be more effective than immediately promoting advanced digital technologies. For firms that are already relatively prepared, more suitable interventions may include incentives for digital integration, supply chain collaboration, and wider innovation partnerships. This differentiated approach is consistent with prior empirical findings that SMIs' transition toward CE and digitalisation is influenced by external support, institutional conditions, and the broader ecosystem surrounding the firm. Accordingly, the framework can help policymakers move from generic support schemes toward more efficient, readiness-based intervention design.

Overall, the managerial implications of this study emphasise that readiness for CE and I4.0 adoption is not merely a technological issue, but a multidimensional and system-oriented challenge shaped by procurement choices, organisational values, human capability, digital infrastructure, and external collaboration. The integration of prioritisation, classification, and subsystem mapping within the proposed framework therefore provides a practical basis for data-driven decision-making and more sustainable transformation pathways for F&B SMIs.

Author Contributions

Conceptualization, Q., D.A.P., S.M.A., and R.D.K.; methodology, Q. and D.A.P.; software, S.M.A. and R.D.K.; validation, Q., D.A.P., S.M.A., and R.D.K.; formal analysis, Q. and D.A.P.; investigation, Z.A.M.R.; resources, Z.A.M.R.; data curation, Z.A.M.R.; writing—original draft preparation, Q., D.A.P., S.M.A., R.D.K., and Z.A.M.R.; writing—review and editing, Q., D.A.P., and S.M.A.; visualization, R.D.K. and Z.A.M.R.; supervision, Q.; project administration, S.M.A.; funding acquisition, Q. All authors have read and agreed to the published version of the manuscript.

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Informed Consent Statement

This study involved humans as respondents to fill the questionnaire (non- interventional study). No experiments or clinical trials were conducted. All participants provided written informed consent prior to engagement in this study.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix

Table A1. Readiness small and medium-sized industries (SMIs) classification

Industry SMIs Code	Readiness Score (CC_i)	Readiness Category	Industry SMIs Code	Readiness Score (CC_i)	Readiness Category	Industry SMIs Code	Readiness Score (CC_i)	Readiness Category
1	0.5732	DR	51	0.4612	DR	101	0.6057	R
2	0.6696	R	52	0.5680	DR	102	0.4147	DR
3	0.5308	DR	53	0.4919	DR	103	0.5982	DR
4	0.6346	R	54	0.4475	DR	104	0.5912	DR
5	0.4165	DR	55	0.6270	R	105	0.6678	R
6	0.6341	R	56	0.5310	DR	106	0.5975	DR
7	0.5192	DR	57	0.6940	R	107	0.5509	DR
8	0.4143	DR	58	0.6228	R	108	0.5136	DR
9	0.5409	DR	59	0.6761	R	109	0.5728	DR
10	0.6550	R	60	0.4764	DR	110	0.4574	DR
11	0.5081	DR	61	0.5126	DR	111	0.5135	DR
12	0.5043	DR	62	0.4345	DR	112	0.6564	R
13	0.3886	ER	63	0.5583	DR	113	0.5392	DR
14	0.5095	DR	64	0.6126	R	114	0.5864	DR
15	0.6199	R	65	0.3972	ER	115	0.4369	DR
16	0.7151	R	66	0.5340	DR	116	0.4822	DR
17	0.5708	DR	67	0.5525	DR	117	0.4431	DR
18	0.5922	DR	68	0.6181	R	118	0.5754	DR
19	0.4492	DR	69	0.5578	DR	119	0.5255	DR
20	0.6126	R	70	0.5639	DR	120	0.5576	DR
21	0.6233	R	71	0.6997	R	121	0.5924	DR
22	0.6852	R	72	0.5703	DR	122	0.5811	DR
23	0.6018	R	73	0.6205	R	123	0.4465	DR
24	0.4277	DR	74	0.5122	DR	124	0.6154	R
25	0.5761	DR	75	0.6595	R	125	0.4134	DR
26	0.6594	R	76	0.7182	R	126	0.4869	DR
27	0.4999	DR	77	0.5819	DR	127	0.4017	DR
28	0.4109	DR	78	0.5430	DR	128	0.6244	R
29	0.4457	DR	79	0.5292	DR	129	0.5165	DR
30	0.6098	R	80	0.5280	DR	130	0.3905	ER
31	0.4258	DR	81	0.4343	DR	131	0.6328	R
32	0.5324	DR	82	0.5296	DR	132	0.6054	R
33	0.3888	ER	83	0.3842	ER	133	0.5749	DR
34	0.3338	ER	84	0.5119	DR	134	0.3793	ER
35	0.4449	DR	85	0.6669	R	135	0.4975	DR
36	0.5127	DR	86	0.5560	DR	136	0.5574	DR

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Table A1 (continued)

Industry SMIs Code	Readiness Score (CC_i)	Readiness Category	Industry SMIs Code	Readiness Score (CC_i)	Readiness Category	SMIs	Readiness Score (CC_i)	Readiness Category
37	0.6357	R	87	0.5239	DR	137	0.4322	DR
38	0.4695	DR	88	0.4488	DR	138	0.5648	DR
39	0.5821	DR	89	0.6166	R	139	0.5598	DR
40	0.7705	R	90	0.3504	ER	140	0.5124	DR
41	0.6125	R	91	0.3502	ER	141	0.6069	R
42	0.5502	DR	92	0.5968	DR	142	0.5621	DR
43	0.4957	DR	93	0.5942	DR	143	0.5586	DR
44	0.5942	DR	94	0.4453	DR	144	0.7159	R
45	0.5263	DR	95	0.4956	DR	145	0.6649	R
46	0.6850	R	96	0.4630	DR	146	0.6641	R
47	0.6657	R	97	0.6379	R	147	0.5963	DR
48	0.7705	R	98	0.5509	DR	148	0.6395	R
49	0.3654	ER	99	0.5983	DR	149	0.7265	R
50	0.6592	R	100	0.5548	DR	150	0.4788	DR